MODULE 4 | AIRCRAFT FLIGHT CONTROLS

ASYS 245 | PART A | ATA 27 | CORNELL NOTES

SECTION 1 • Axes + Primary Controls

CUES / QUESTIONS

1. Which three axes does an aircraft rotate around?

2. Which control surface causes roll, and how?

3. What is the elevator and what motion does it control?

4. How does elevator deflection affect aircraft pitch?

5. What is the rudder and how does it control yaw?

NOTES

1 | THREE AXES OF FLIGHT



AXIS | MOTION

LONGITUDINAL
Roll • nose to tail

LATERAL
Pitch • wing to wing

VERTICAL
Yaw • top to bottom

The axes pass through the aircraft CG (centre of gravity).

2 | ROLL: AILERONS (LONGITUDINAL AXIS)



Ailerons move in opposite directions. A down-deflected aileron increases local lift; the opposite up aileron reduces lift. The difference rolls the aircraft.

Outer trailing edges of both wings

AILERON DOWN → MORE LIFT

AILERON UP → LESS LIFT

3 | PITCH: ELEVATOR (LATERAL AXIS)

Elevator: hinged control surface on the horizontal stabilizer (empennage).

Moving the elevator changes tail aerodynamic force and pitches the nose up or down.

Elevator up → nose up



4 | YAW: RUDDER (VERTICAL AXIS)

Rudder: movable surface on the vertical stabilizer at the tail.

Rudder deflection produces a side force on the tail, causing the nose to yaw about the vertical axis.

Rudder right → nose yaws right



SECTION 1 | STUDENT SUMMARY

- Roll is about the longitudinal axis; ailerons control it.
- Yaw is about the vertical axis; the rudder controls it.
- Pitch is about the lateral axis; the elevator controls it.
- Opposing surface forces rotate the airplane about its CG.